DO NOW — RECALL

Year 13 autumn mock: Paper 2

Name: _____ Class: _____ Date: _____

1. What makes a program's output testable?

2. What is a precondition?

3. Which part of a partition is guaranteed to be in its final position?

4. Push 4,9,2 onto a stack; pop once; enqueue the popped value to queue [7,8], then dequeue. Give remaining stack and queue.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Defined success criteria and unambiguous rules.

2. Model answer

A requirement that must hold before an algorithm runs.

3. Model answer

The pivot (under the specified partition method).

4. Model answer

Stack bottom-to-top [4,9]; pop gives 2; queue [7,8,2] then dequeue 7 leaves [8,2].

Vocabulary — know these terms

base case — Condition ending recursion.

trace table — Record intermediate algorithm state.

Big O — Asymptotic algorithm growth.

auxiliary space — Extra storage used by an algorithm.

NEW LEARNING

Read, annotate and explain

Answer original practice, not OCR past paper

This diagnostic samples programming and algorithms, including learning from earlier Year 12 lessons.

Time management

Allow roughly 15 minutes per 15 marks.

Method precision

Binary search must shrink bounds, sorting traces must show complete passes, recursion must include a terminating condition.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Unmarked warm-up: trace binary search midpoint for 7 in [1,3,5,7,9].

Predict the method

Check the middle 5 and discard indexes 0–2 because target 7 is larger.

Check and explain

Next middle is index 3 containing 7, so return 3.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

(a) State the key precondition for binary search and explain why it is necessary. [2] (b) Trace binary search for 11 in [1,3,5,7,9,11,13], giving low/high/mid at each step and resulting index. [7] (c) Compare worst-case complexities of linear and binary searches. [6]

Question 2

(a) Starting with [5,1,4,2], show the array after ascending bubble-sort Pass 1, Pass 2 and a final no-swap checking Pass 3. [9] (b) Compare bubble sort with merge sort in worst-case time complexity and auxiliary space. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

(a) Write recursive C# function SumTo(n) returning 1+…+n for n≥0, with base case. [8] (b) Trace SumTo(3) showing the recursive calls and returned value. [5] (c) State one drawback of recursive solutions. [2]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A correctly written recursive function without a base case is guaranteed to terminate.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.